B. Rebuttal Figures (ICML 2026)

Table 17. **Training datasets used for instruction finetuning of Llama-3.2-1B.**

Overlap Method	Training Data
Task splitting method (1)	`pinzhenchen/alpaca-cleaned-de`, `pinzhenchen/alpaca-cleaned-fr`, `DanielSc4/alpaca-cleaned-italian`, `ai4bharat/indic-instruct-data-v0.1` (Hindi split), `pinzhenchen/alpaca-cleaned-es`, `airesearch/WangchanThaiInstruct`
Dataset splitting	`ai4bharat/indic-instruct-data-v0.1` (English split)

B.1. Perplexity of CPT/finetuned LLMs

Table 18. **Perplexity before and after tuning.** *Full-parameter CPT, dataset overlap (tuned on the `Tiny-Codes` dataset). Model: Llama-3.2-1B-Instruct.*

Overlap	0	0.25	0.5	0.75	1
Pretrained	5.9180	5.9180	5.9180	5.9180	5.9180
Finetuned	2.3411	2.3395	2.3399	2.3398	2.3410

Table 19. **Perplexity before and after tuning.** *LoRA CPT, dataset overlap (tuned on the `Tiny-Codes` dataset). Model: Llama-3.2-1B-Instruct.*

Overlap	0	0.2	0.4	0.6	0.8	1
Pretrained	5.9180	5.9180	5.9180	5.9180	5.9180	5.9180
Finetuned	2.5886	2.5889	2.5888	2.5881	2.5891	2.5883

Table 20. **Perplexity before and after tuning.** *Full-parameter CPT, task overlap method (1). Model: Llama-3.2-1B-Instruct. We use the datasets specified in Table 2.*

Overlap	0	0.33333	0.66666	1
Pretrained	16.6426	20.1345	18.6461	14.9055
Finetuned	13.2929	18.6845	19.4327	15.6712

Table 21. **Perplexity before and after tuning.** *LoRA CPT, task overlap method (1). Model: Llama-3.2-1B-Instruct. We use the datasets specified in Table 2.*

Overlap	0	0.33333	0.66666	1
Pretrained	16.6426	20.1345	18.6461	14.9055
Finetuned	11.3460	14.3676	13.6794	10.7242

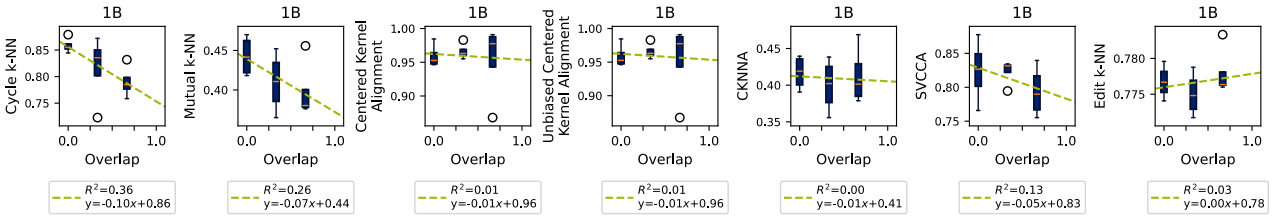
Table 22. **Perplexity before and after tuning.** *Full-parameter instruction finetuning, dataset overlap. Model: Llama-3.2-1B. We use the datasets specified in Table 17.*

Overlap	0	0.25	0.5	0.75	1
Pretrained	7.7537	7.7537	7.7537	7.7537	7.7537
Finetuned	2.7398	2.7645	2.7200	2.7395	2.7325

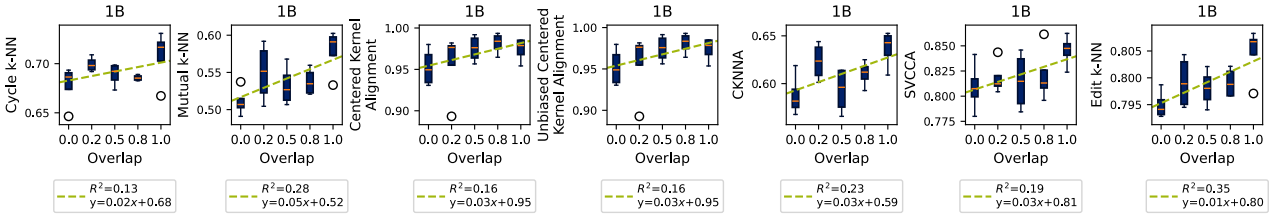
Table 23. **Perplexity before and after tuning.** *Full-parameter instruction finetuning, task overlap method (1). Model: Llama-3.2-1B. We use the datasets specified in Table 17.*

Overlap	0	0.33333	0.66666	1
Pretrained	8.4613	7.0020	8.2017	8.6175
Finetuned	2.6459	2.5639	2.6194	2.8027

B.2. Instruction-finetuning of Llama-3.2-1B



(a) Task splitting.



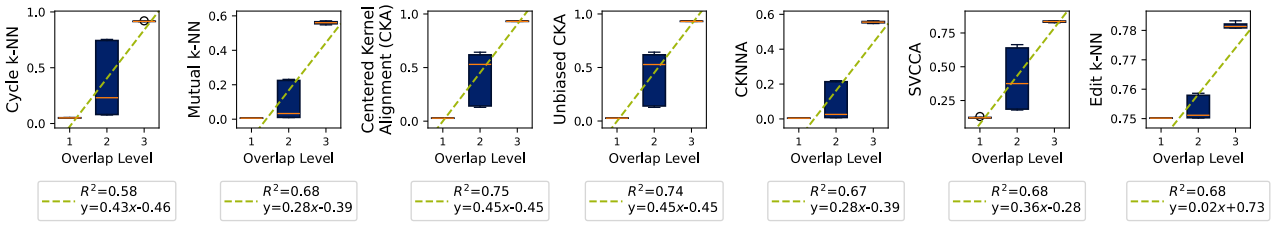
(b) Dataset splitting

Figure 30. **All similarity scores for task and dataset splitting (full-parameter instruction finetuning, Llama-3.2-1B)**. *Representational similarity measured using metrics in Huh et al. (2024) for Llama-3.2-1B under task splitting method (1) and dataset splitting with error bars representing deviation across different splitting and training seeds. Evaluated on validation set. See 17 for a summary of the datasets used.*

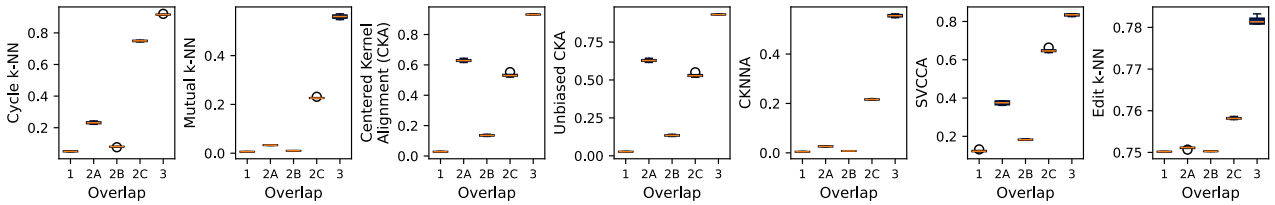
B.3. Language classification, task-overlap method (2)

Table 24. **Accuracy and number of classes across overlap settings for task overlap method (2)**. *Shows accuracy on language classification as well as the labels used for each overlap setting. Used the* `buruzaemon/amazon_reviews_multi` *dataset, where each data point has three labels: product-category, stars rating, and language. We finetuned from the* `FacebookAI/xlm-roberta-base` *model. Note: the last two columns correspond to the set of paired-up models used to compute representational similarity and are trained in exactly the same way other than the random states.*

Overlap	0%	0%	50%	50%	50%	100%	100%
Label	Product-Category	Stars	Stars + Language	Stars + Product-Category	Language + Product-Category	All Labels (split 0)	All Labels (split 1)
Mean Accuracy	0.548	0.627	0.625	0.335	0.543	0.326	0.325
Number of Classes	31	5	30	155	186	930	930



(a) Aggregated results with line of regression



(b) Disaggregated results

Figure 31. **All similarity scores for task overlap method (2) (full-parameter instruction finetuning, XLM-RoBERTa-base)**. *Representational similarity measured using metrics in Huh et al. (2024) for XLM-RoBERTa-base under task splitting method (2) with error bars representing deviation across different splitting and training seeds. Evaluated on validation set. Showing both disaggregated and aggregated results.*